

CONTACT MANAGEMENT SYSTEM

1. Project Overview

Description

The Contact Management System is a menu-driven Python application developed using Functions and Dictionaries. It allows users to manage contacts by adding, searching, updating, deleting, and viewing contact details. The application stores all contact information permanently in a JSON file and also provides the option to export data into CSV format.

The project demonstrates the practical use of Python programming concepts including dictionaries, functions, file handling, validation, exception handling, and modular programming.

2. Project Objectives

The objectives of this project are:

- Learn Python Functions
- Learn Dictionary Data Structure
- Perform CRUD Operations
- Implement Input Validation
- Store Data Permanently
- Export Data to CSV
- Practice Modular Programming
- Improve Problem Solving Skills

3. Technologies Used

Technology	Purpose
Python 3	Programming Language
JSON	Permanent Storage
CSV	Export Contacts
re Module	Email & Phone Validation
datetime	Timestamp Management

os Module	File Handling
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4. Project Structure

week3-contact-manager/

|— contacts_manager.py

|— contacts_data.json

|— contacts_export.csv

|— test_contacts.py

|— README.md

|— requirements.txt

|— .gitignore

5. Installation & Setup

Step 1

Clone the repository

```
git clone < https://github.com/mohit-214/week3\_contact\_manager\_system/tree/main/contact%20management%20System >
```

Step 2

Move to project folder

```
cd week3-contact-manager
```

Step 3

Install dependencies

```
pip install -r requirements.txt
```

Step 4

Run the project

```
py "c:/Users/Mohit/contact management System/Contact Management System.py"
```

6. Code Structure Explanation

The project is divided into multiple reusable functions.

Each function performs only one task.

This improves

- Readability
- Reusability
- Maintainability

Major functions include:

- `add_contact()`
- `search_contacts()`
- `update_contact()`
- `delete_contact()`
- `save_contacts()`
- `load_contacts()`
- `export_to_csv()`
- `view_statistics()`

7. Features

- ✓ Add Contact
- ✓ Search Contact
- ✓ Update Contact
- ✓ Delete Contact
- ✓ Display Contacts
- ✓ JSON Storage
- ✓ CSV Export
- ✓ Contact Statistics
- ✓ Phone Validation
- ✓ Email Validation
- ✓ User Friendly Menu
- ✓ Error Handling

8. Function Explanation

`validate_phone()`

Purpose

Validates phone numbers.

Checks

- Only digits
 - Length between 10–15 digits
-

validate_email()

Purpose

Checks whether the email format is valid using Regular Expressions.

add_contact()

Purpose

Adds a new contact after validating all inputs.

Stores

- Name
 - Phone
 - Email
 - Address
 - Group
 - Date Created
 - Updated Date
-

search_contacts()

Purpose

Searches contacts using partial matching.

update_contact()

Purpose

Updates existing contact details.

delete_contact()

Purpose

Deletes selected contact after confirmation.

save_contacts()

Purpose

Stores dictionary data into JSON.

load_contacts()

Purpose

Loads contacts from JSON file.

export_to_csv()

Purpose

Exports contacts into CSV format.

view_statistics()

Purpose

Displays

- Total Contacts
 - Group-wise Contacts
 - Recently Updated Contacts
-

9. Input Validation

The application validates

✓ Contact Name

✓ Phone Number

✓ Email Address

This prevents invalid data from entering the system.

10. Error Handling

The project handles

- Missing JSON File
 - Invalid Phone Number
 - Invalid Email
 - Empty Name
 - Invalid Menu Choice
 - JSON Decode Errors
-

11. JSON Data Storage

All contacts are stored inside

contacts_data.json

Each contact contains

- Phone
 - Email
 - Address
 - Group
 - Created Time
 - Updated Time
-

12. CSV Export

Users can export all contacts into

contacts_export.csv

This allows data to be opened in

- Excel

- Google Sheets
- LibreOffice

13. Contact Statistics

The application displays

- Total Contacts
- Contacts by Group
- Recently Updated Contacts

This provides a quick summary of stored data.

JSON Data Persistence

Contacts are automatically saved after every change.

Example:

```
PS C:\Users\Mohit\contant management System> py "c:/Users/Mohit/contant management System/Contact Manag

=====
                MAIN MENU
=====
1. Add New Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. View All Contacts
6. Export to CSV
7. View Statistics
8. Exit
Enter your choice (1-8): 5

--- ALL CONTACTS (5 total) ---
=====
👤 Mohit Bhandari
📞 9899754125
✉️ mohitbhandari@gmail.com
📍 Haryana , faridabad
👥 family
-----
👤 Shaziya Malik
📞 78545967458
✉️ shaziya@gmail.com
📍 Dehradun
👥 Family
```

```

0 contacts_data.json > {} Sachin
1 {
2     "Mohit Bhandari": {
3         "phone": "9899754125",
4         "email": "mohitbhandari@gmail.com",
5         "address": "Haryana , faridabad",
6         "group": "family",
7         "created_at": "2026-07-02T13:19:12.943688",
8         "updated_at": "2026-07-02T13:19:12.943738"
9     },
10    "Shaziya Malik": {
11        "phone": "78545967458",
12        "email": "shaziya@gmail.com",
13        "address": "Dehradun",
14        "group": "Family",
15        "created_at": "2026-07-02T13:20:00.655352",
16        "updated_at": "2026-07-02T13:20:00.655367"
17    },

```

Advantages:

- Lightweight
- Human-readable
- Easy to edit
- Fast loading

14. Sample Output Screenshot

Add Contact

```

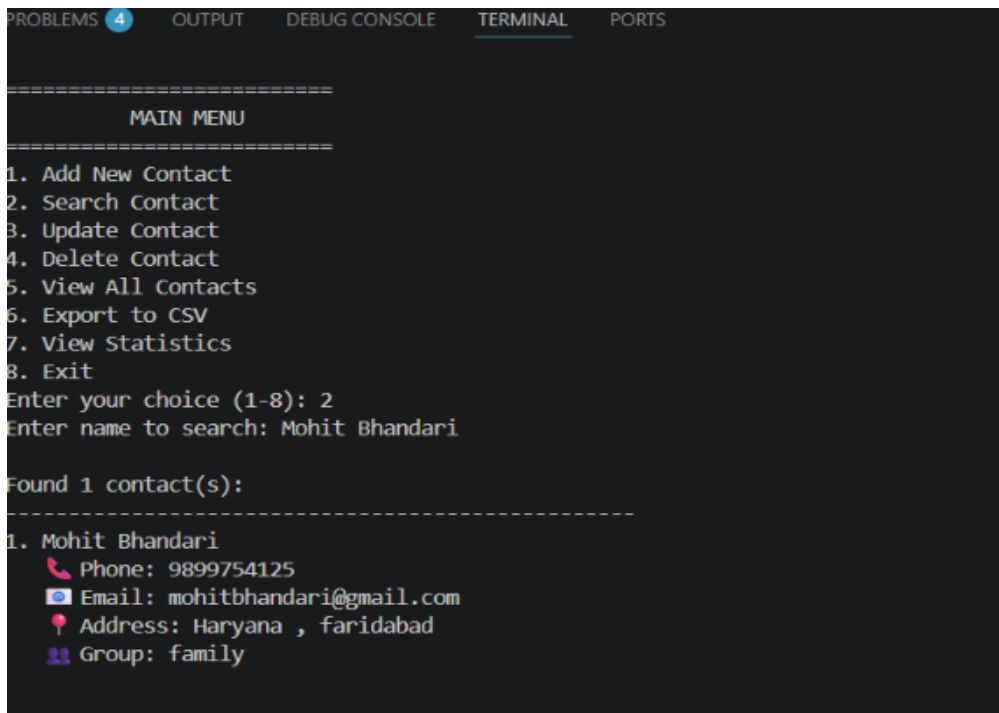
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
MAIN MENU
=====
1. Add New Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. View All Contacts
6. Export to CSV
7. View Statistics
8. Exit
Enter your choice (1-8): 1

--- ADD NEW CONTACT ---
Enter contact name: Rohit
Enter phone number: 9568425136
Enter email (optional, press Enter to skip): rohit@gmail.com
Enter address (optional): Delhi uttam nagar
Enter group (Friends/Work/Family/Other): other
✅ Contact 'Rohit' added successfully!
✅ Contacts saved to contacts_data.json

```


Search Contact



```
=====
MAIN MENU
=====
1. Add New Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. View All Contacts
6. Export to CSV
7. View Statistics
8. Exit
Enter your choice (1-8): 2
Enter name to search: Mohit Bhandari

Found 1 contact(s):
-----
1. Mohit Bhandari
  Phone: 9899754125
  Email: mohitbhandari@gmail.com
  Address: Haryana , faridabad
  Group: family
```

15. CSV Export

The application exports contacts into:

contacts.csv

Columns:

- Name
- Phone
- Email
- Address
- Group

16. Challenges & Solutions

Challenge 1

Duplicate contact names.

Solution

Implemented duplicate checking before saving.

Challenge 2

Phone validation.

Solution

Created validation logic supporting multiple formats.

Challenge 3

Searching efficiently.

Solution

Used case-insensitive partial matching with `lower()`.

Challenge 4

Saving data permanently.

Solution

Used JSON with automatic saving after every operation.

Challenge 5

Handling file errors.

Solution

Implemented try-except blocks to prevent program crashes.

Author

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Project: Contact Management System

Week 3